

ZHAW – Zurich University of Applied Sciences

School of Engineering

# Drone Control Tuning

Development of an Offline Betaflight Drone Control Tuner

---

**Autor:** Yuri Bianchi, Dario Jurietti and Janick Dort  
**Modul / PA:** Project Thesis  
**Supervisor:** Michael Ernst Peter  
**Co-Supervisor:** Dr. Prof. Ruprecht Altenburger

Winterthur, 2. Dezember 2025

# 1 Abstract

This project focuses on the development of a method for offline tuning of rate controllers for FPV racing drones based on the open-source firmware Betaflight. The quality of the rate control loop has a direct influence on flight stability, the suppression of disturbances such as prop wash, and the agility of the drone. Until now, rate controller tuning has mostly been performed through time-consuming trial and error.

The aim of the project is to develop a modular, open-source solution for offline tuning based on an existing proof of concept. The code will be implemented in MATLAB, accompanied by a lightweight Python version, and supplemented with detailed Markdown documentation and examples on GitHub. The project aims to provide the community with a simple way to efficiently optimise drone parameters and thereby improve flight performance.

The results will be published in a public GitHub repository and documented in a report. In addition to the software development, the project also includes practical work with FPV drones to validate the developed methods.

# 2 Foreword

This document summarizes the results of our project thesis, which we worked on during the fall semester 2025 at the ZHAW School of Engineering. For us as systems engineering students, this project provided an ideal opportunity to apply advanced control theory and signal processing methods to a real, highly dynamic system. FPV drones present an interesting engineering challenge due to their highly dynamic behaviour and the need for precise control. This motivated us to investigate mathematical modelling, control engineering aspects, and offline tuning methods aimed at improving flight performance.

Our motivation was to translate the theoretical control concepts we learned over the past three years to a real, highly dynamic system. We also aimed to develop a tool that not only meets engineering standards but also provides real value to the FPV community by replacing subjective tuning with objective data analysis.

This work was carried out at the Institute of Mechatronic Systems (IMS) at the ZHAW School of Engineering. We would like to express our gratitude to our supervisor, Michael Peter, for his guidance and support throughout this project. His expertise, constructive feedback, and willingness to take the time to assist us in developing complex algorithms were invaluable.

## Inhaltsverzeichnis

|                                  |          |
|----------------------------------|----------|
| <b>1 Abstract</b>                | <b>2</b> |
| <b>2 Foreword</b>                | <b>2</b> |
| <b>3 Introduction</b>            | <b>4</b> |
| 3.1 Initial situation . . . . .  | 4        |
| 3.2 Task and Objective . . . . . | 4        |
| <b>4 Preparation</b>             | <b>5</b> |
| 4.1 Chirp . . . . .              | 5        |
| 4.1.1 Chirp Concept . . . . .    | 5        |

## **3 Introduction**

Text...

### **3.1 Initial situation**

Text...

### **3.2 Task and Objective**

Text...

## 4 Preparation

### 4.1 Chirp

Text...

#### 4.1.1 Chirp Concept

Text..